

ANANTAARAA

Detailed explanation of the proposed solution

Anantaaraa is a digital platform that makes India's ancient astronomical instruments interactive and accessible, blending immersive mixed reality with modern science and classical wisdom.

It is a **cross-platform app** that offers support for both **IOS and Android** as well as **website support**.

Requirements and Features

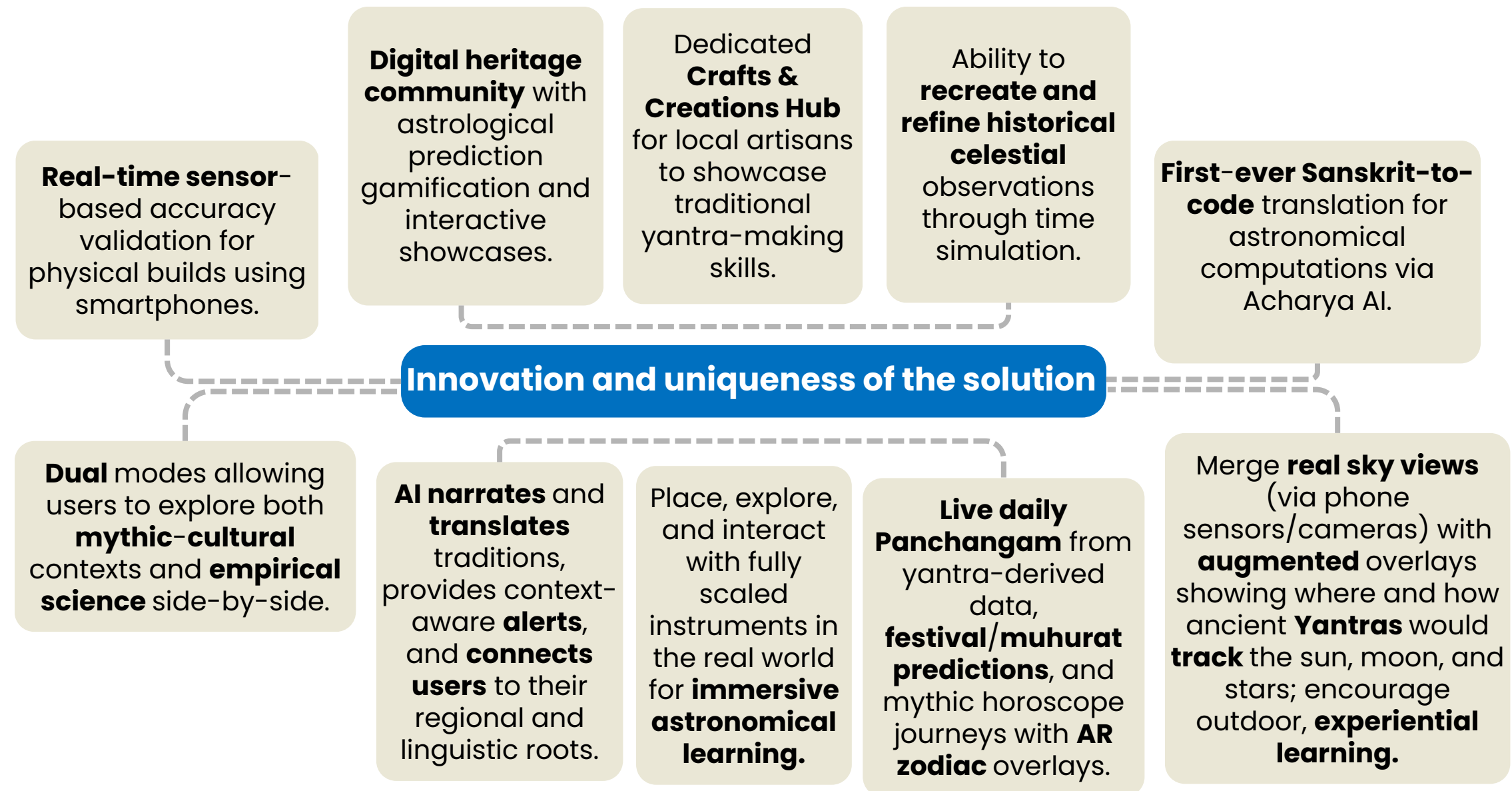
- **Dynamic computation** of yantra geometry for any Indian latitude/longitude.
- Support for multiple **historical instruments'** designs.
- **Geodetic calibration** from Ujjain prime meridian to modern coordinates.
- Astronomical **event prediction** blending ancient and modern techniques.
- Integration of **Panchangam, festival dates, and Sanskrit cosmology**.
- Validation mechanism for physical builds using **mobile sensors**.

Novel Features:

- Immersive **AR/VR** and **holographic** heritage experiences.
- **Community** tools for sharing builds, hosting virtual exhibitions, and competitions.
- **AI-powered** real-time **voice** guidance for **observations** and **measurements**.
- **Location-based** cultural **notifications** for auspicious and seasonal events.
- Dual-mode experience: **Heritage** Explorer & **Scientist** Mode.
- **Revenue** model with freemium, **premium** features, **institutional** licensing, and **heritage** partnerships.

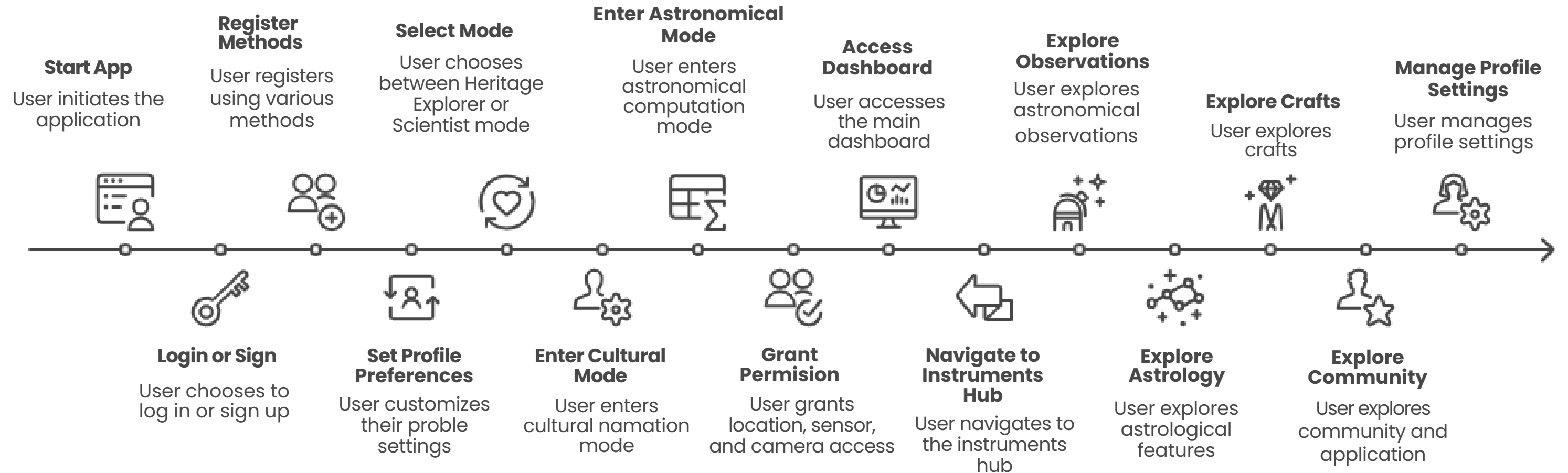
How it addresses the problem

- Automates **complex** mathematical derivations for diverse **geographies**.
- Combines **ancient** and **modern** astronomy into a unified, interactive system.
- Simplifies building through **CAD** exports and in-app construction **accuracy** checks.
- Engages both **cultural** audiences and **scientific** communities effectively.
- Allows **academic** validation and **correction** of historical observational data.

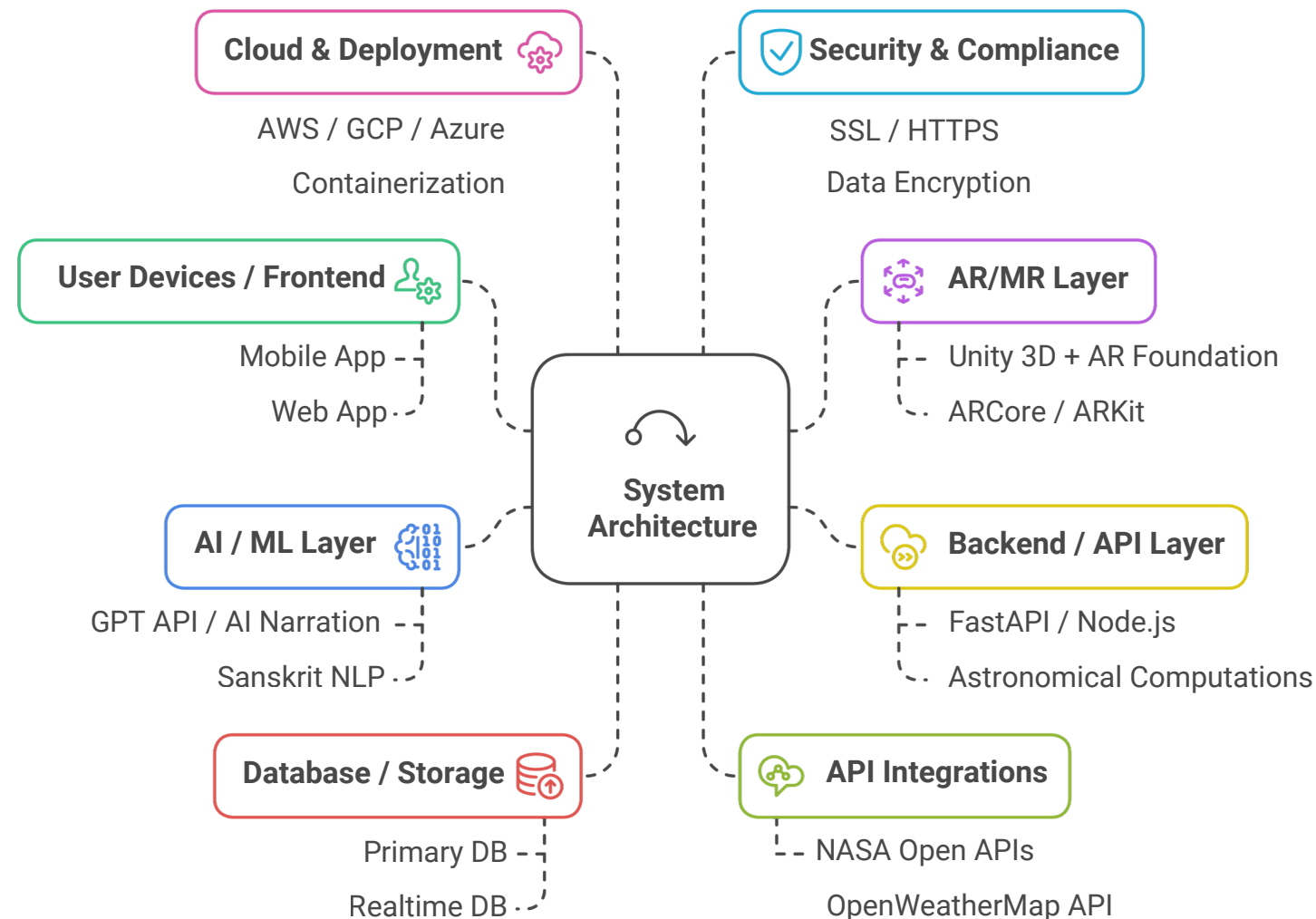


TECHNICAL APPROACH

Flow chart of the Whole process



Technologies



Working Prototype Link :

- Youtube link for the working prototype of "ANANTAARAA"
- https://youtu.be/dGhiZ5Q_VIU

FEASIBILITY AND VIABILITY

Feasibility of the idea

Technological Feasibility

- Leverages existing AR, AI, geodetic, and mobile tech for accurate yantra simulation and interaction.

Operational Feasibility

- Compatibility and integration of new technology.
- Tools, equipment, and production method appropriateness for the project scope.
- Regular content and AI updates

Financial Feasibility

- Freemium and premium models
- Institutional licensing potential
- Government grants accessible
- Artisan marketplace revenue stream

Social & Cultural Feasibility

- Multilingual inclusive support
- Cultural authenticity prioritized
- Respect for astrological traditions
- Heritage awareness promotion

Strategies for Overcoming Challenges

- **Offline Capabilities:** Enable offline mode and edge computing to handle connectivity issues.
- **Build agile, modular architecture** with phased development to manage technical complexity.
- **Engage cultural experts** and local communities continuously for **data validation and sensitivity reviews**.
- **Design Inclusive UX:** Build **multilingual, accessible, and intuitive interfaces** for diverse users.
- **Regular Content Audits:** Periodically **review and update content** with expert panels to maintain accuracy and respect.
- **Adaptive Learning Algorithms:** **Continuously improve AI models** based on user interactions and data trends.

Potential Challenges and Risks

Cultural Authenticity:

Balancing scientific accuracy with faithful cultural representation is complex.

Advanced Features:

Implementing AR, 3D modeling, and celestial simulations requires technical expertise and resources.

User Diversity:

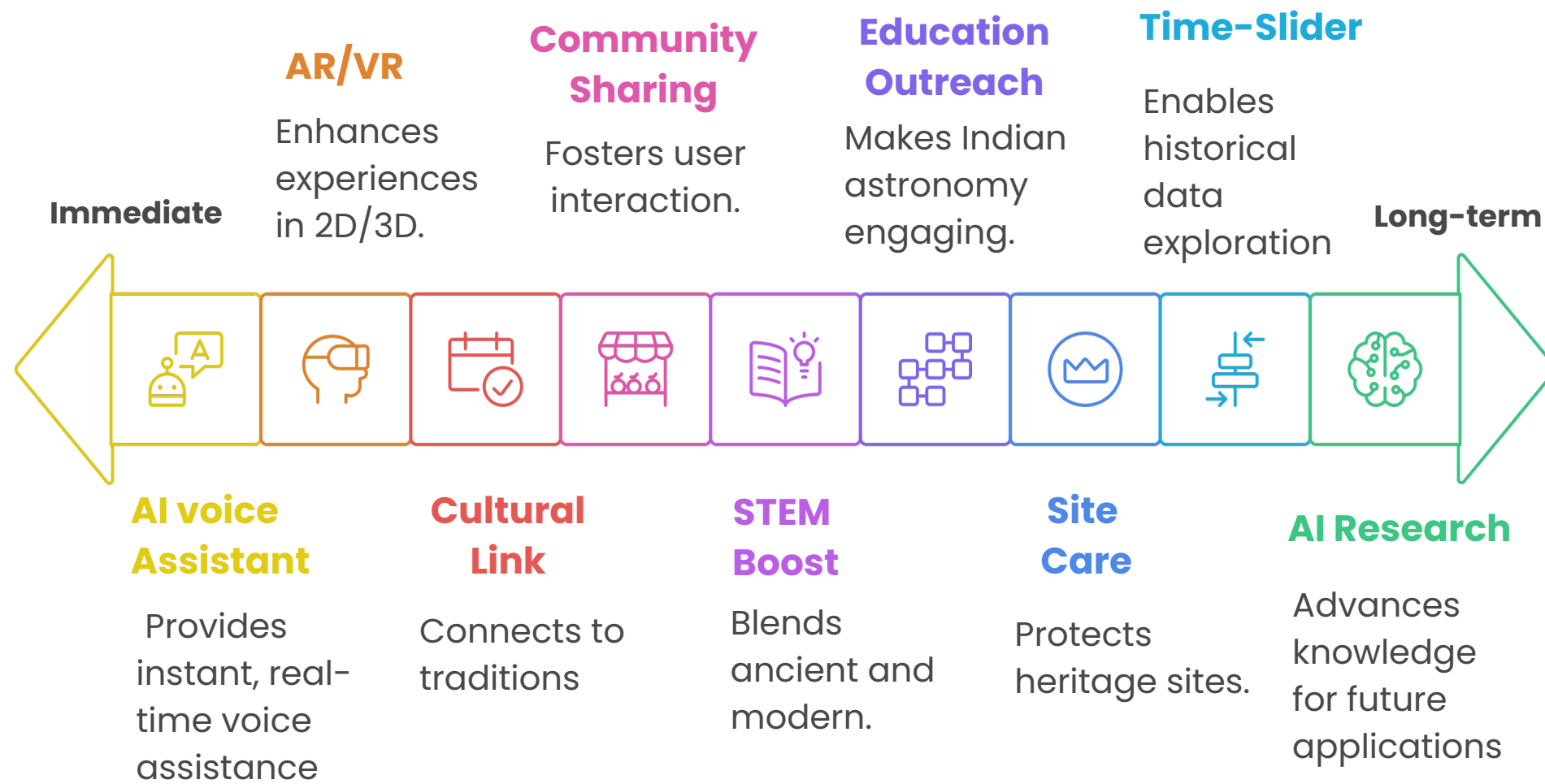
Designing an intuitive interface that fits both heritage explorers and scientific users' needs.

Data Privacy:

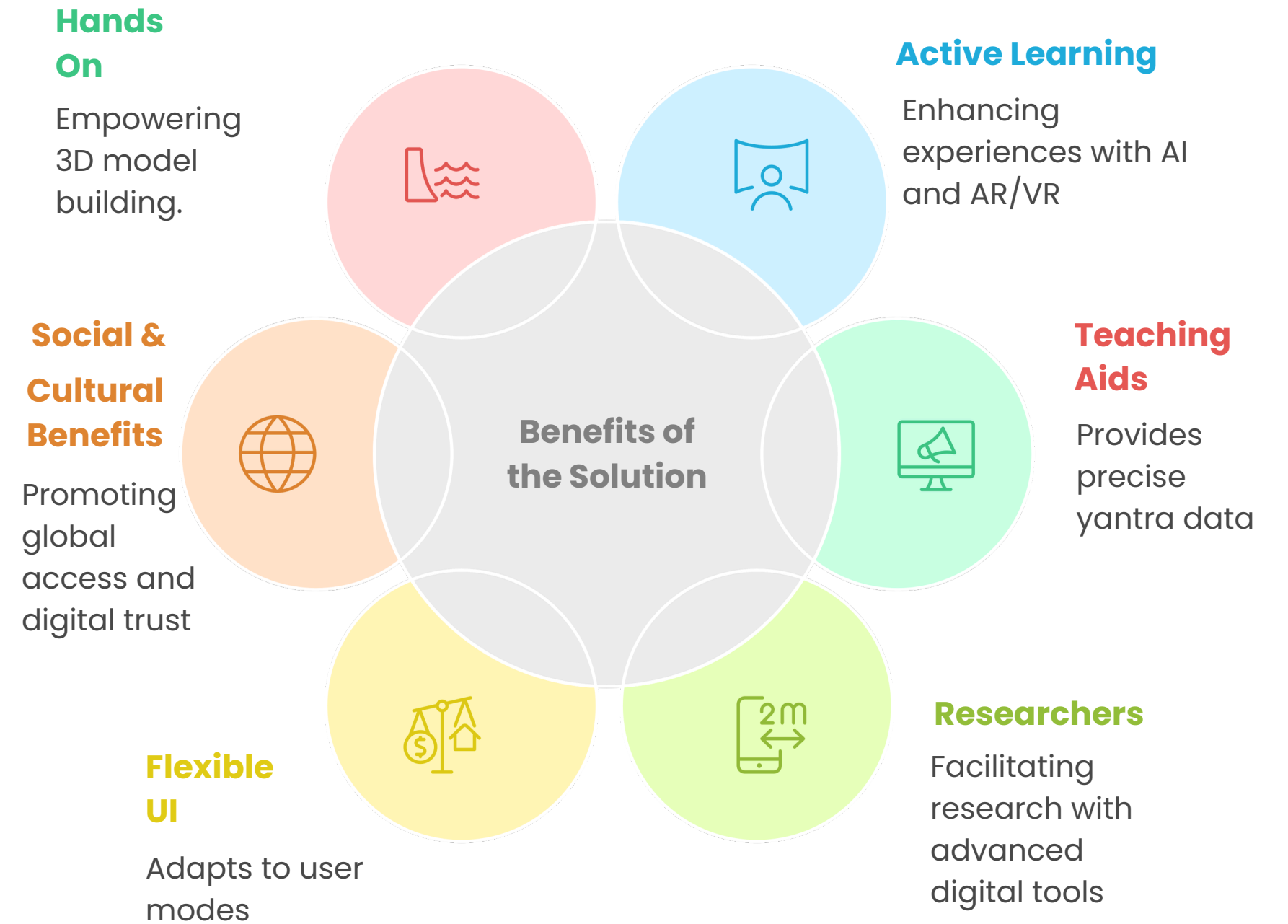
Protecting user location and collaborative data against breaches is essential.

IMPACT AND BENEFITS

Understanding impacts : from immediate to long-term effects



Benefits of the Solution



RESEARCH AND REFERENCES



District Ujjain Official Website

- Observatory Information
- <https://ujjain.nic.in/en/tourist-place/observatory/>



Indian Astronomical Instruments and Jantar Mantar Studies

- Sawai Jai Singh II and Observatories research
- <https://ui.adsabs.harvard.edu/abs/2015hae..book.2017J/abstract>



Jantar Mantar Instruments Overview

- Jantar Mantar official website
- <https://www.jantarmentar.org/learn/observatories/instruments/samrat/index.html>



Ancient Indian Astronomy and Instrument Construction

- BharatPunj Astronomy overview
- <https://bharatpunj.com/wiki/indian-astronomy/>



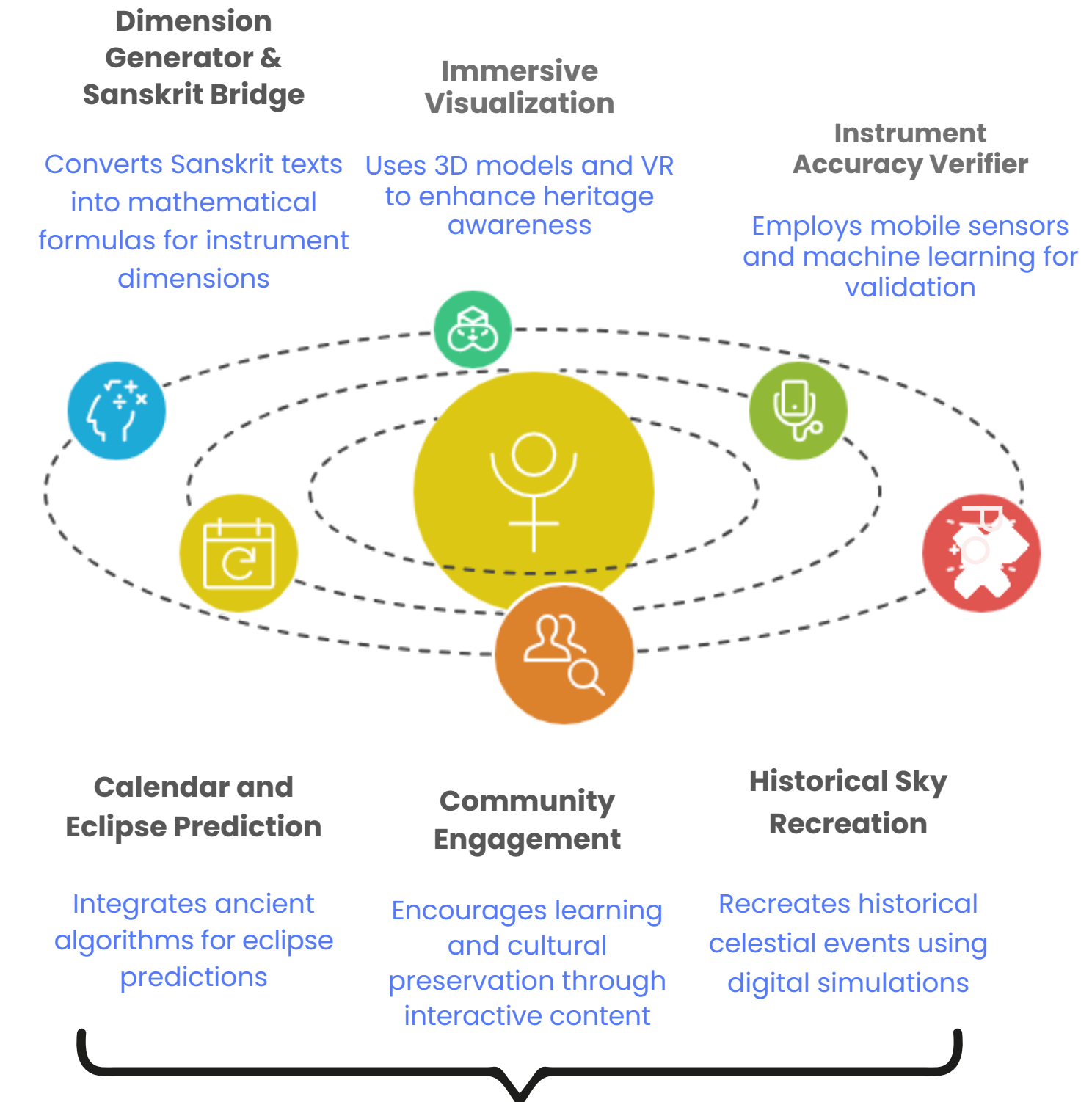
Techniques of Determining Latitude in Indian Astronomy

- S. K. Kulkarni (2024)
- <https://www.anantaajournal.com/archives/2024/vol10issue6/PartA/10-5-44-848.pdf>



Historical Meridian and Timekeeping in Ujjain

- 5 Senses Tours, Ujjain Meridian study
- <https://5sensestours.com/ujjain-unveiling-the-ancient-prime-meridian-of-the-world/>



Tap on the text to open the reference link